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TECHNOLOGICAL UNIVERSITY DUBLIN
CITY CAMPUS - GRANGEGORMAN

TU856 – B.Sc.(Hons) Computer Science
TU857 – B.Sc.(Hons) Computer Science (Infrastructure)
TU858 – B.Sc.(Hons) Computer Science International

Year 3

SEMESTER 1 EXAMINATIONS 2024/25

CMPU3010 Databases 2

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Exam Duration: 2 hours

Instructions: Answer 2 out of 3 questions. All questions carry equal weight.

1.

- (a) You are given the following SQL query and its query plan. The query is executed on the *dvdrental* database. Please see Appendix A for a database schema for this database.

SQL query:

```
SELECT
    c.customer_id,
    (SELECT COUNT(*) FROM rental r
     WHERE r.customer_id = c.customer_id)
    AS total_rentals
FROM
    customer c
ORDER BY
    total_rentals DESC;
```

Query Plan:

```
Sort (cost=650.00..660.00 rows=1000 width=16)
  Sort Key: (SubPlan 1)
  -> Seq Scan on customer c (cost=0.00..600.00
    rows=1000 width=8)
    SubPlan 1
      -> Aggregate (cost=0.40..0.41 rows=1
        width=8)
        -> Seq Scan on rental r
          (cost=0.00..300.00 rows=1000 width=8)
          Filter:
            (r.customer_id = c.customer_id)
Planning Time: 0.250 ms
Execution Time: 72.356 ms
```

- (i) Describe the inefficiencies in this query plan and explain why they contribute to the query's cost. **(10 marks)**
- (ii) Recommend a more efficient way to structure the SQL of this query to reduce redundant processing. **(10 marks)**
- (iii) Explain how adding an index could improve the performance of this query and reduce the sorting and filtering costs. **(5 marks)**
- (b) Describe two real-world scenarios where the flexibility of application logic might outweigh the benefits of stored procedures? **(10 marks)**

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(c) Consider the following scenario:

A new e-commerce platform needs to design a MongoDB database schema to handle product listings, customer orders, and user reviews. The platform's requirements include:

1. **Product Listings:**

- a. Store details such as `product_id`, `name`, `description`, `price`, `category`, and `available_stock`.
- b. Each product should also have a list of reviews, with each review containing `review_id`, `user_id`, `rating`, and `comment`.
- c. Support efficient queries for products by `category` and within specific price ranges.

2. **Customer Orders:**

- a. Store information for each order, including `order_id`, `user_id`, `order_date`, `items`, and `total_cost`.
- b. Each order item should include `product_id`, `quantity`, and `price_at_purchase`.
- c. Support efficient queries to retrieve all orders by a specific user and detailed views of individual orders.

3. **User Reviews:**

- a. Allow users to retrieve all reviews they have posted.
- b. Enable efficient queries to retrieve reviews for individual products.

Additionally, assume that as the platform grows, **new fields or data structures may be introduced**. The schema should allow for these adjustments without requiring a rigid structure.

(i) MongoDB's schema is flexible and doesn't require a fixed structure. Given this flexibility:

- Describe **how you would structure each of the three main entities** (Product Listings, Customer Orders, and User Reviews) in **json** while keeping future schema changes in mind.
- Specify any **embedding** or **referencing** strategies for handling relationships, justifying each decision with the need to balance performance and flexibility.

(10 marks)

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- (ii) MongoDB supports both embedding and referencing of documents.
- Discuss one advantage and one disadvantage of using **embedding** for product reviews within Product Listings.
 - Provide an example where switching to referencing might be beneficial.
- (5 marks)

2. (a) Study the case outlined below and answer the questions that follow.

Grange Car Engineering (GCE) provides car servicing and repair services in North County Dublin. The shop uses a ledger to write in details of their customers, cars, jobs, and parts used in repairs.

The customer contacts GCE to request a service or a repair. The customer's name is written in the book along with details of the car – make, model and reg - and the next available date is given to the customer for the job.

As the mechanic works on the car, details of the parts used are written in the book. Note that customers can have more than one car.

The volume of business is increasing and the company want to automate their administration.

A snapshot of the ledger used is shown in Appendix B *table 1*.

Use this to do the following tasks, explaining what is meant by each level of Normal Form:-

- (i) Represent the data in Unnormalised Form. (6 marks)
- (ii) Represent the data in First Normal Form. (6 marks)
- (iii) Represent the data in Second Normal Form. (6 marks)
- (iv) Represent the data in Third Normal Form. (6 marks)
- (v) Draw a fully normalized ERD to represent the entities derived by the normalisation process, showing primary and foreign keys, attributes and relationships. (6 marks)

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- (b) GCE also provides car fleet services to its customers and has been using a mobile Customer Relationship Manager app to capture and record information on each customer. For this service, GCE undertakes to fix any issues identified by its customer by a specified response time. The app's data model is illustrated in Appendix B *figure 2*.

Write SQL for each of the following queries:-

- (i) Assuming only one contract per customer, list the number of issues per service contract. **(5 marks)**
 - (ii) How many customers had issues resolved by the top ranked mechanic (resolvedBy) **(5 marks)**
 - (iii) For the top 10 issue names, list 1) their position in the top 10, 2) the issue name and 3) how many times it has occurred. **(5 marks)**
- Your answer should generate 10 rows or less in the output.
- (iv) Which customers have raised all the issue types? **(5 marks)**

3. (a) (i) What are windows functions in Postgres SQL? What are windows function suitable for? **(4 marks)**
- (ii) Explain the difference between **sequential scan** and **index scan** in POSTGRES query explain plan. For what type of queries is the index scan a much faster solution? Provide an example. **(5 marks)**
- (iii) Consider the following table in POSTGRES

```
create table persons(  
    person_id integer,  
    person_name varchar(20),  
    person_surname varchar(20),  
    person_age integer);
```

We create an index on the table as follows:

```
create index p_age on persons(person_age);
```

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We are interested in counting the number of persons by age.
We run the following query:

```
select person_age, count(person_age) from persons where
person_age < 20 group by person_age;
```

And the explain analyze command for this query gives the following output:

	QUERY PLAN
1	GroupAggregate (cost=0.29..9.31 rows=75 width=12)
2	Group Key: person_age
3	-> Index Only Scan using p_age on persons (cost=0.29..7.61 rows=190 width=4)
4	Index Cond: (person_age < 20)

Explain the above output, including the total cost of the query and the cost of each sub-operation and the meaning of Index Only, Index Cond. and Group Aggregate.

(5 marks)

- (b) (i) Describe the concept of transactions deadlock in databases, providing an example.
- (ii) Describe a strategy to resolve the problem of transactions deadlock in databases
- (iii) Explain the meaning of the built-in *new* and *old* variables in a Postgres trigger, providing an example of their usage and explaining if the two variables can be used in a before insert trigger.
- (c) You are required to design a database to store information about employees and their managerial relationships. Each employee is described by an employee_id, an employee name and a salary. There is a relationship among the employees: employee *a* is linked to employee *b* if *a* is *b*'s manager. The date employee *a* became *b*'s manager is also stored.

(6 marks)

(5 marks)

(5 marks)

You are required to:

- (i) Provide a relational schema to show how to store employees and the "manager" relationship. Provide tables, fields and show primary and foreign keys.
- (ii) Write an SQL query to get all the list of employees directly managed by the employee with id=3

(6 marks)

(3 marks)

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- (iii) Write an SQL query to get all the list of employees managed by employee with id=3 directly or managed by people managed by employee with id=3. **(5 marks)**
- (iv) Provide a *json* structure to store the same information provided in the relational model into a *mongodb* collection. **(6 marks)**

Appendix A

dvdrental Database Schema

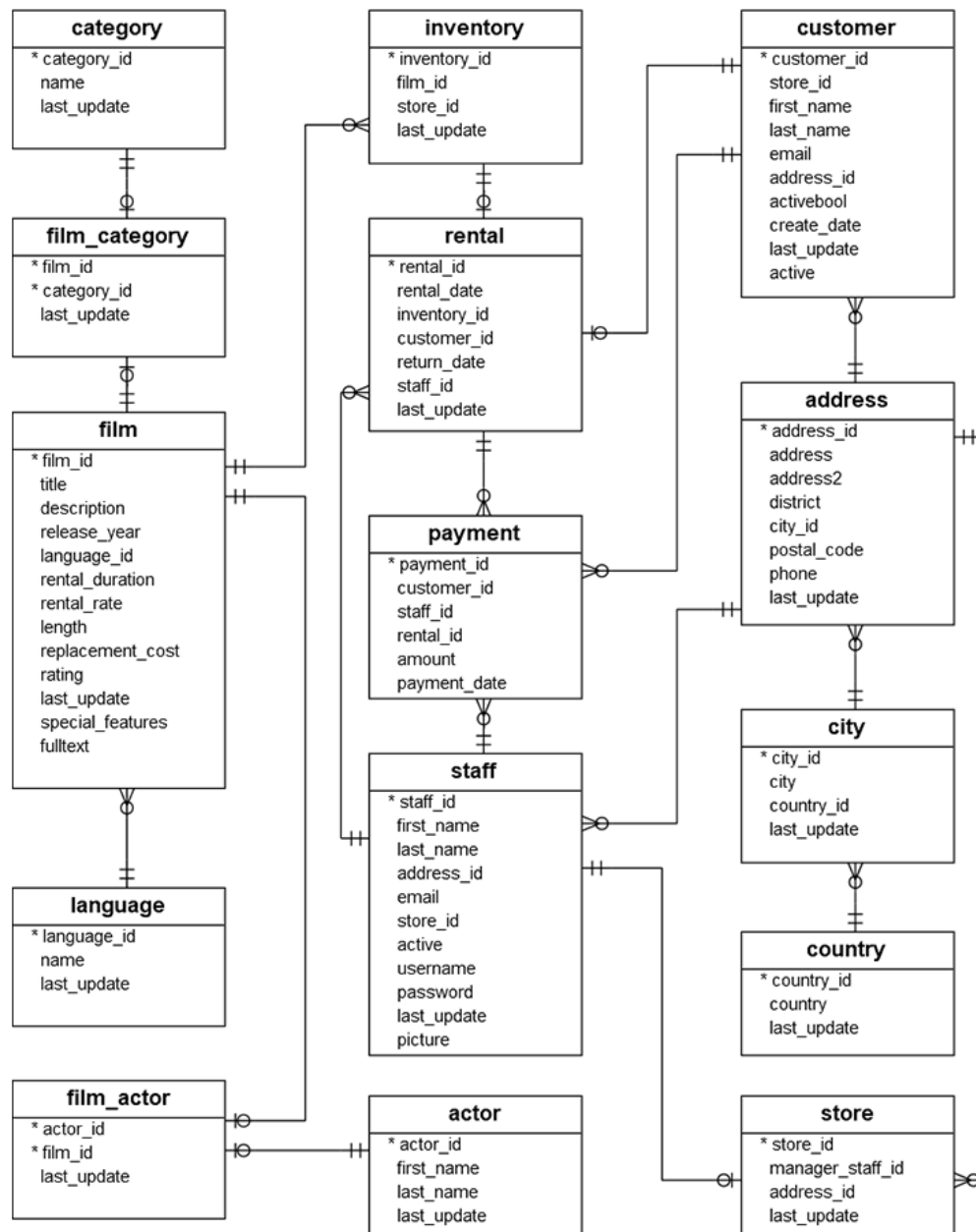


Figure 1 dvdrental database schema

Appendix B

Cust-omer ID	Cust-omer Name	Car Make	Model	Registration	Job#	Repair Date	Repair Description	Mechanic ID	Part ID	Part Name	Part Cost	VAT	Total
001	DT	Kia	Sportage	10SO1234	1	01/05/2024	Service	WE1	101	Oil Filter	10	2.30	12.30
001	DT	Kia	Sportage	10SO1234	1	01/05/2024	Service	WE1	102	Air Filter	5	1.15	6.15
001	DT	Kia	Sportage	10SO1234	1	01/05/2024	Service	WE1	103	5L Oil	50	11.50	61.50
001	DT	Ford	Model 1	02WX1234	2	01/04/2024	Service	WH	191	2L Oil	50	11.50	61.50
002	CH	Opel	Corsa	96D1234	3	01/05/2024	Repair – brake failure	WE2	104	Brake Pad	60	13.80	73.80
002	CH	Opel	Corsa	96D1234	3	01/05/2024	Repair – brake failure	WE2	105	Brake Fluid	20	4.60	24.60
002	CH	Opel	Corsa	96D1234	3	01/05/2024	Repair – front lamp	WE2	105	Brake Fluid	10	2.30	12.30

Table 1 Car Workshop Ledger

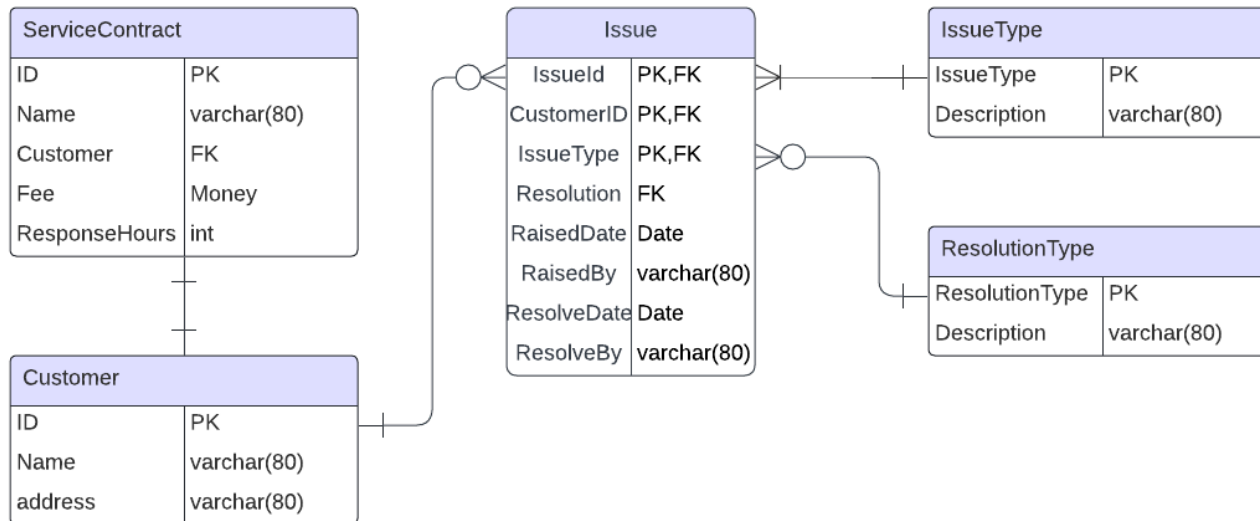


Figure 2 - Customer Relationship Management – Database schema